SE LAB-5

PES2UG24CS102
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1. Pylint



2. Flake8



3. Bandit

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(.venv) andy@andy-Lenovo-V15-G2-ALC-Ua:/media/andy/7CF6C4F2F6C4AE28/Users/Andy/Desktop/COLLEGE/SE/lab5/L5_36_smart_home$ python -m bandit -r smart_home.py
[main] INFO profile include tests: None
[main] INFO profile exclude tests: None
[main] INFO cli include tests: None
[main] INFO cli exclude tests: None
[main] INFO running on Python 3.12.3
Run started:2026-10-14 18:26:28.473633+00:00

Test results:
>> Issue: [B404:blacklist] Consider possible security implications associated with the subprocess module.
   Severity: Low   Confidence: High
   CWE: CWE-78 (https://cwe.mitre.org/data/definitions/78.html)
   More Info: https://bandit.readthedocs.io/en/1.9.4/blacklists/blacklist_imports.html#b404-import-subprocess
   Location: ./smart_home.py:4:0
3 import json
4 import subprocess
5 from pathlib import Path

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>> Issue: [B607:start_process_with_partial_path] Starting a process with a partial executable path
   Severity: Low   Confidence: High
   CWE: CWE-78 (https://cwe.mitre.org/data/definitions/78.html)
   More Info: https://bandit.readthedocs.io/en/1.9.4/plugins/b607_start_process_with_partial_path.html
   Location: ./smart_home.py:89:17
88      try:
89          result = subprocess.run(
90              ["bash", str(scene_path)], check=False, shell=False
91          )
92      except OSError as error:

--------------------------------------------------
>> Issue: [B603:subprocess_without_shell_equals_true] subprocess call - check for execution of untrusted input.
   Severity: Low   Confidence: High
   CWE: CWE-78 (https://cwe.mitre.org/data/definitions/78.html)
   More Info: https://bandit.readthedocs.io/en/1.9.4/plugins/b603_subprocess_without_shell_equals_true.html
   Location: ./smart_home.py:90:17
89          result = subprocess.run(
90              ["bash", str(scene_path)], check=False, shell=False
91          )
92      except OSError as error:
93          print("Unable to run scene:", error)

--------------------------------------------------

Code scanned:
 Total lines of code: 92
 Total lines skipped (#nosec): 0

Run metrics:
 Total issues (by severity):
  Undefined: 0
  Low: 3
  Medium: 0
  High: 0
 Total issues (by confidence):
  Undefined: 0
  Low: 0
  Medium: 0
  High: 3
Files skipped (0):
(.venv) andy@andy-Lenovo-V15-G2-ALC-Ua:/media/andy/7CF6C4F2F6C4AE28/Users/Andy/Desktop/COLLEGE/SE/lab5/L5_36_smart_home$
```